Math 105L **Lecture 2-1**

Power Functions, Monomial Ordering, & Polynomials

1. Power Functions

A **power function** is any function of the form:

$$\mathbf{f(x) = kx^p}$$

where the coefficient k and exponent p are real constants.

- **Radicals as Fractional Powers:** $\sqrt{x} = \mathbf{x^{1/2}}, \quad \sqrt[3]{x} = \mathbf{x^{1/3}}$.
- **Reciprocals as Negative Powers:** $\frac{1}{\sqrt[3]{x^7}} = \frac{1}{x^{7/3}} = \mathbf{x^{-7/3}}$.

Relative Ordering & Geometry of Monomials x^n $(n \geq 1)$ All integer monomial curves $y = x^n$ pass through the common points $(\mathbf{0}, \mathbf{0})$ and $(\mathbf{1}, \mathbf{1})$ in the first quadrant:

- **Inside the Unit Interval** $(0, 1)$**:** Smaller powers are **strictly greater** than higher powers because multiplying numbers between 0 and 1 reduces their size:

$$\mathbf{x > x^2 > x^3 > x^4 > x^5} \quad \text{for } \mathbf{x} \in (\mathbf{0}, \mathbf{1})$$

- **Beyond the Unit Interval** $(1, \infty)$**:** Higher powers grow much faster and **dominate**:

$$\mathbf{x^5 > x^4 > x^3 > x^2 > x} \quad \text{for } \mathbf{x} \in (\mathbf{1}, \infty)$$

- **Symmetry Across the Axes:**
  - **Even Powers** (x^2, x^4)**:** Even functions $(f(-x) = f(x))$, symmetric across the y-axis. Both arms point up $(+\infty$ as $x \to \pm\infty)$.
  - **Odd Powers** (x, x^3, x^5)**:** Odd functions $(f(-x) = -f(x))$, symmetric through the origin. Passes through $(-1, -1)$.

2. Polynomials & Degree Classification

Definition of a Polynomial:
A **polynomial** is a sum of power functions with non-negative integer exponents:

$$\mathbf{p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0} \quad (\mathbf{a_n \neq 0})$$

The largest exponent n is the **degree** of the polynomial:

$n = 1$:	**Linear**	$(y = mx + b)$	$n = 2$:	**Quadratic**	$(ax^2 + bx + c)$
$n = 3$:	**Cubic**	$(ax^3 + \dots)$	$n = 4$:	**Quartic**	$(ax^4 + \dots)$
$n = 5$:	**Quintic**	$(ax^5 + \dots)$	$n \geq 6$:	**Degree-n**	

End Behavior Dominance: As $x \to \pm\infty$, the highest degree term $a_n x^n$ vastly outgrows all lower terms combined, determining the ultimate end-behavior of the polynomial.